WORKED SOLUTION

STEP II 2021, Question 3

Pure • Floor and Fractional Parts

STEP II 2021, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Every real number splits uniquely into an integer part and a fractional part in $[0,1)$. So when an equation says "an integer plus a number in $[0,1)$ equals 4.9", both pieces are forced: the integer must be 4 and the fraction must be 0.9. In (ii) and (iii) the equations are tangled, but adding all three lets every $\lfloor t \rfloor + \{t\}$ recombine into t. That gives one clean linear equation, and subtracting each original equation from it separates the unknowns into exactly that "integer plus fraction" form.

Solution

Throughout we use: if n is an integer, $0 \le f < 1$ and $n + f = c$, then $n = \lfloor c \rfloor$ and $f = \{c\}$. This holds because $n \le c < n+1$, which pins down n as the largest integer not exceeding c.

Part (i)

In the first equation $\lfloor x \rfloor$ is an integer and $0 \le \{y\} < 1$, so

$$\lfloor x \rfloor = 4, \qquad \{y\} = 0.9.$$

In the second, $\lfloor y \rfloor$ is an integer and $0 \le \{x\} < 1$, so $\lfloor y \rfloor = \lfloor -1.4 \rfloor = -2$ and $\{x\} = -1.4 - (-2) = 0.6$. Hence

$$x = 4 + 0.6 = 4.6, \qquad y = -2 + 0.9 = -1.1.$$

These satisfy both equations ($4 + 0.9 = 4.9$ and $0.6 - 2 = -1.4$), and we have shown they are the only solution.

Part (ii)

Add the three equations. Each of x, y, z appears once on its own, and once more split as $\lfloor\cdot\rfloor + \{\cdot\}$ across two different equations:

$$(x+y+z) + (\{x\} + \lfloor x\rfloor) + (\lfloor y\rfloor + \{y\}) + (\{z\} + \lfloor z\rfloor) = 3.9 + 5.3 + 5,$$

that is, $2(x+y+z) = 14.2$, so

$$x + y + z = 7.1.$$

Ask yourself Now that we know $x+y+z$, what is left over when we take away one of the original equations?

Subtracting the first equation from $x+y+z = 7.1$ removes x and leaves $y - \lfloor y\rfloor$ and $z - \{z\}$:

$$\{y\} + \lfloor z\rfloor = 7.1 - 3.9 = 3.2,$$

which is the required result. In the same way, subtracting the second and third equations gives

$$\lfloor x\rfloor + \{z\} = 7.1 - 5.3 = 1.8, \qquad \{x\} + \lfloor y\rfloor = 7.1 - 5 = 2.1.$$

Each of these is "integer plus fractional part", so

$$\begin{aligned} \lfloor z\rfloor &= 3, & \{y\} &= 0.2,\\ \lfloor x\rfloor &= 1, & \{z\} &= 0.8,\\ \lfloor y\rfloor &= 2, & \{x\} &= 0.1. \end{aligned}$$

Therefore $x = 1.1$, $y = 2.2$, $z = 3.8$. This is the only possible solution, and it does work:

$$1.1 + 2 + 0.8 = 3.9, \qquad 0.1 + 2.2 + 3 = 5.3, \qquad 1 + 0.2 + 3.8 = 5.$$

Part (iii)

The same trick works. Adding the three equations, with $2\lfloor y\rfloor + 2\{y\} = 2y$,

$$2x + 4y + 2z = 14.2, \qquad \text{so} \qquad x + 2y + z = 7.1.$$

Subtracting each equation in turn from this gives

$$2\{y\} + \lfloor z\rfloor = 3.2, \qquad \lfloor x\rfloor + \{z\} = 1.8, \qquad \{x\} + 2\lfloor y\rfloor = 2.1.$$

The middle equation is of the familiar type, so $\lfloor x\rfloor = 1$ and $\{z\} = 0.8$.

In the third equation, $2\lfloor y\rfloor$ is still an integer and $0 \le \{x\} < 1$, so the earlier reasoning applies: $2\lfloor y\rfloor = 2$ and $\{x\} = 0.1$. Hence $\lfloor y\rfloor = 1$ and $x = 1.1$.

Ask yourself Does the first equation still force a unique answer?

No. Here $2\{y\}$ lies in $[0, 2)$, not $[0, 1)$, so there are two ways to write 3.2 as an integer plus a number in $[0, 2)$:

- $\lfloor z \rfloor = 3$, $2\{y\} = 0.2$, so $\{y\} = 0.1$;
- $\lfloor z \rfloor = 2$, $2\{y\} = 1.2$, so $\{y\} = 0.6$.

(If $\lfloor z \rfloor \geq 4$ then $2\{y\} < 0$, and if $\lfloor z \rfloor \leq 1$ then $2\{y\} \geq 2.2$; both are impossible.) With $\lfloor y \rfloor = 1$ and $\{z\} = 0.8$, this gives two candidates:

$$(x, y, z) = (1.1,\ 1.1,\ 3.8) \qquad \text{or} \qquad (x, y, z) = (1.1,\ 1.6,\ 2.8).$$

Both satisfy the original system:

$$\begin{aligned} (1.1, 1.1, 3.8): &\quad 1.1 + 2 + 0.8 = 3.9, \quad 0.1 + 2.2 + 3 = 5.3, \quad 1 + 0.2 + 3.8 = 5; \\ (1.1, 1.6, 2.8): &\quad 1.1 + 2 + 0.8 = 3.9, \quad 0.1 + 3.2 + 2 = 5.3, \quad 1 + 1.2 + 2.8 = 5. \end{aligned}$$

So the system in (iii) has exactly these two solutions.

Checks

Substitute the answers back in, as above: every equation balances. As an independent check, a computer search with x and y running over all multiples of 0.1 in $[-10, 10]$ found exactly one solution to (i), one to (ii) and exactly the two above for (iii). Note that $\{y\} + \lfloor z \rfloor = 0.2 + 3 = 3.2$ for the solution of (ii), as it should.

Common slips

- Getting the floor of a negative number wrong: $\lfloor -1.4 \rfloor = -2$, not -1, so $\{x\} = 0.6$ in (i).
- In (iii), assuming $2\{y\}$ is a fractional part and so finding only one solution. It lies in $[0, 2)$, which is why a second solution appears.
- Not checking the candidates in the original equations. The deductions above only show they are the only possibilities.

Beyond STEP

Writing $x = \lfloor x \rfloor + \{x\}$ is the same as working modulo 1: the fractional part is the image of x in the circle $\mathbb{R}/\mathbb{Z}$. Floor and fractional parts are everywhere in number theory, for example in Legendre's formula for the power of a prime dividing $n!$, and in the study of how the fractional parts of $n\alpha$ spread out over $[0, 1)$ for irrational α. Davenport's *The Higher Arithmetic* is a lovely introduction to this circle of ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.